

True Chaos, Monte Carlo, and Uncertainty Geometry

A Reproducible Scale-Comparison Battery

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Abstract

Monte Carlo quantifies variation under a declared sampling law. It does not, by itself, decide whether the reported law is forced by an experiment or inherited from the declaration. We extend the uncertainty-geometry battery with a direct comparison across chaotic, periodic, quasi-periodic, and singular observation models. The tests combine finite-time Lyapunov slopes, total variation, Wasserstein distance, entropy spread, observation-scale sweeps, ordinary $N^{-1/2}$ Monte Carlo convergence, and coverage at a fold wall. All five registered checks pass for seed 20261010. The decisive result is not that geometric computation always differs from Monte Carlo. They match after a mixing frontier on observable quotient quantities. Geometry detects additional structure exactly where Monte Carlo convergence alone is silent: persistence of input laws, resolution-dependent distinguishability, non-identifiable branches, and the difference between a small sampling error and a forced answer.

1 Question and relation to the earlier uncertainty battery

The earlier paper *RG - Uncertainty Geometry* separated uncertainty constrained by data from uncertainty inherited along observational fibres. Its UG02 and UG10 checks already showed that chaos may forget a declared ensemble while periodic dynamics may retain it, and that Monte Carlo is objective only for outputs on which admissible declarations agree. The present battery sharpens that comparison in two ways. First, forgetting is tested simultaneously in several probability metrics and across observation scales. Second, ordinary Monte Carlo convergence is placed next to a deliberately conditional periodic output. Both estimates exhibit the textbook sampling slope, but only one is declaration-insensitive.

The claim is therefore deliberately limited:

Uncertainty geometry does not replace Monte Carlo as a numerical integrator. It supplies a structural certificate telling us when the Monte Carlo target is forced, conditional, or set-valued.

2 Geometric criterion

Let an experiment be a measurable map $F : X \rightarrow Y$ together with an admissible family $\mathcal{A}$ of input laws on X . For a requested output functional T , define the propagated answers

$$\mathcal{T} = \{T(F_{\#}\mu) : \mu \in \mathcal{A}\}.$$

At observation resolution ε and scale s , let d_s be a task-appropriate statistical distance.

Definition 1 (Forced and conditional output). *The output is forced at scale s when*

$$\text{diam}_{d_s}(\mathcal{T}) \leq \varepsilon_s.$$

It is conditional when the diameter remains resolved. At a singular wall, the output may instead be a correspondence, namely the full preimage of an observational confidence set.

This criterion is external to Monte Carlo sample size. Increasing the number of draws reduces integration error for each fixed μ , but does not reduce the diameter over $\mathcal{A}$.

Theorem 1 (Monte Carlo rate is not a non-circularity certificate). *Suppose $\hat{T}_N(\mu)$ is a regular Monte Carlo estimator for each fixed $\mu \in \mathcal{A}$, with $\sqrt{N}(\hat{T}_N(\mu) - T(\mu)) \Rightarrow N(0, \sigma_\mu^2)$. Then standard errors may decay as $N^{-1/2}$ for every admissible declaration while $\text{diam}\{T(\mu) : \mu \in \mathcal{A}\} > 0$ remains unchanged.*

Proof. The central limit statement is pointwise in μ . It controls $\hat{T}_N(\mu) - T(\mu)$, not $T(\mu) - T(\nu)$ for distinct admissible laws. Hence sampling uncertainty vanishes while declaration dependence can remain finite. The periodic involution in Section 6 realizes this separation numerically. $\square$

3 Registered systems and diagnostics

The code tests four one-dimensional dynamics: the logistic map at $r = 4$, the tent map, an irrational rotation, and the periodic involution $x \mapsto 1 - x$. A contracting periodic logistic case at $r = 3.5$ is retained only as a negative-Lyapunov control in the local divergence panel. The fold experiment observes $y = b^2 + \epsilon$.

For each applicable system the battery records:

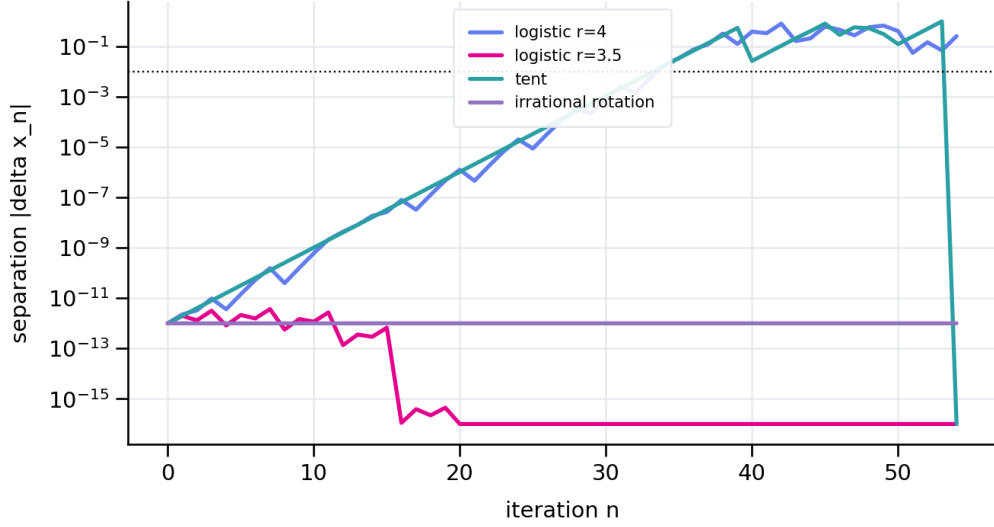
1. finite-separation slope and orbit-average Lyapunov exponent;
2. maximum pairwise total variation, W_1 , and histogram-entropy spread over uniform, Gaussian, and bimodal input ensembles;
3. the same input-law comparison over 8 to 512 histogram bins;
4. replicated Monte Carlo standard errors over $N = 250$ to 16000;
5. Wald coverage versus coverage of the full geometric preimage at a fold.

4 True-chaos slope and the forecast frontier

The logistic map at $r = 4$ has measured orbit-average Lyapunov exponent 0.693467, while the finite perturbation fit gives 0.682453, close to $\log 2 = 0.693147$. The tent control has exponent $\log 2$, the irrational rotation has exponent zero, and the $r = 3.5$ logistic control has exponent -0.872507 . Thus the battery does not identify “chaos” from visually irregular samples or from Monte Carlo variance. It registers exponential separation relative to explicit controls.

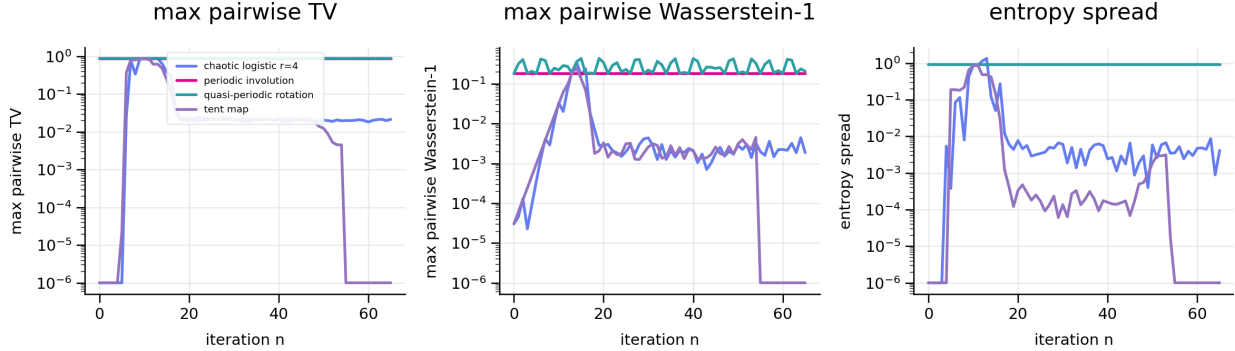
The local slope is not yet a probability statement. Probability becomes forced only when propagated ensembles forget their admissible input differences at the requested observation scale.

Finite perturbations: exponential growth is system-specific



5 Multi-metric and multi-scale forgetting

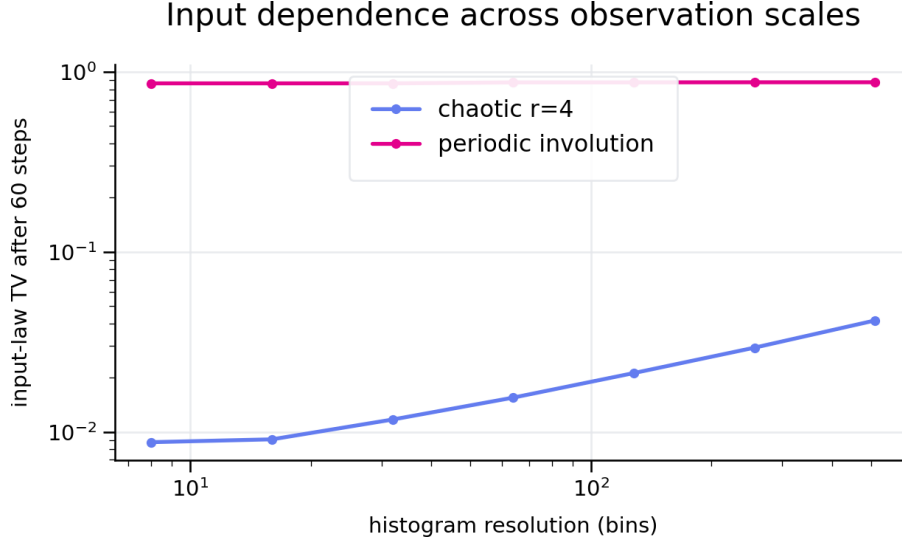
After the frontier, the chaotic logistic ensembles agree to late-window maximum TV 0.02143, $W_1 = 0.00442$, and entropy spread 0.00861. The periodic involution and irrational rotation retain resolved input-law differences. The tent map reaches exact floating-point coalescence in this implementation, so its late zeros should be read as a finite-precision limitation rather than as an exact theorem about real arithmetic.



At coarse-graining from 8 through 512 bins, the chaotic logistic TV ranges from 0.00877 to 0.04160. The periodic involution remains between 0.86339 and 0.87537. This is the scale comparison required for a geometric claim: agreement at one arbitrary binning would be insufficient.

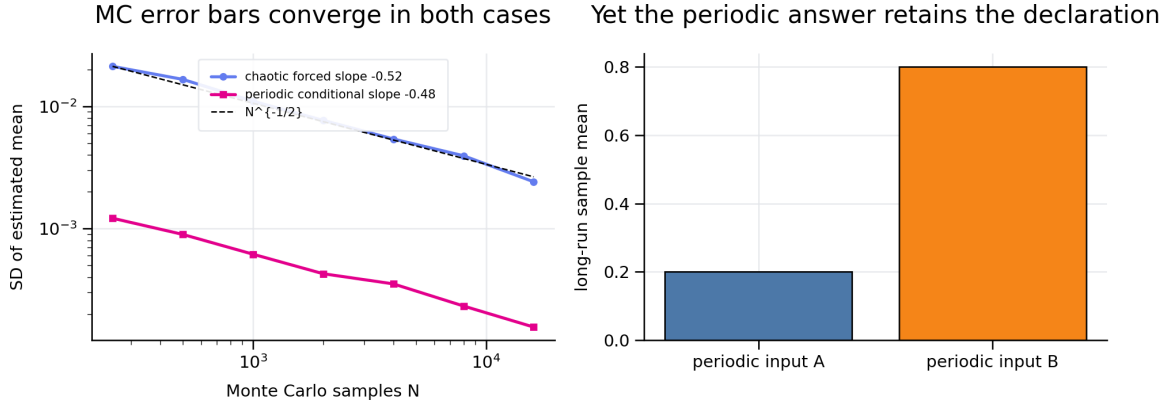
Proposition 1 (Registered mixing certificate). *For the declared ensemble family, sample size, scale range, and seed, the $r = 4$ logistic output is forced within the registered tolerances after the frontier, whereas the periodic involution is conditional.*

Proof. The claim is the conjunction of the registered inequalities in TV over time and over the scale sweep. It is a computational certificate for the declared battery, not a substitute for an analytic mixing theorem. $\square$



6 Where Monte Carlo matches and where it misses structure

For the post-burn-in chaotic mean, the replicated Monte Carlo standard-error slope is -0.5206 . For the periodic involution under one fixed input declaration, it is -0.4843 . Both are consistent with $N^{-1/2}$ convergence. Nevertheless, changing the periodic input declaration moves the limiting mean by 0.60009 . Ordinary Monte Carlo therefore reports a precise conditional answer in the second case.

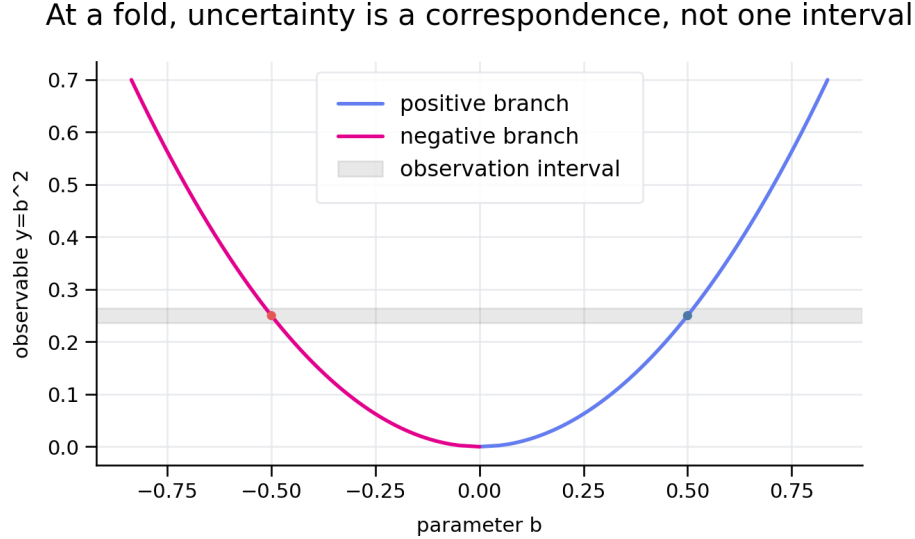


This gives the clean comparison:

Question	Monte Carlo	Uncertainty geometry
Integration error for a fixed law	Estimates it directly	Uses or imports the estimate
Whether the law is data-forced	Does not decide	Tests diameter over admissible laws
Forecast frontier	Can display convergence	Registers when input dependence falls below resolution
Kernel or branch ambiguity	Samples whichever declaration/branch is encoded	Reports fibre dependence or full correspondence
Scale robustness	Requires an explicit sweep	Treats scale as part of the experiment

7 Singular wall: interval versus correspondence

In the fold model $y = b^2 + \epsilon$, the two parameters b and $-b$ are observationally indistinguishable. A positive-branch Wald procedure has coverage 0 for the true value $b = -0.5$ in 5000 replications. Pulling the observational interval back through the full map produces two branches and coverage 0.9534.



Corollary 1. *At regular, identifiable, post-mixing points, uncertainty geometry and Monte Carlo can agree numerically. At a fold, along an observational kernel, or before mixing, the geometric output contains structural information that a pointwise Monte Carlo calculation need not represent.*

8 What has been proved and what has not

Locked by analysis. The theorem above proves that an $N^{-1/2}$ sampling rate cannot certify independence from the input declaration. The fold map proves that branch selection can destroy coverage even when the observational noise model is regular. The input-family diameter is the appropriate direct object for distinguishing forced from conditional outputs.

Locked by the reproducible battery. All five registered checks pass for seed 20261010: positive-Lyapunov discrimination, post-frontier forgetting, scale-robust separation from a non-mixing control, failure of Monte Carlo rate as a structural certificate, and nominal geometric coverage at the fold.

Not claimed. The experiment does not prove mixing for every chaotic system, does not establish a universal numerical tolerance, and does not imply that positive Lyapunov exponent alone forces an invariant probability law. The tent-map late zeros are affected by finite precision. The battery also does not replace an effective-sample-size or dependence diagnostic for Monte Carlo. It adds a logically prior question: is the target itself invariant over admissible declarations?

9 Operational rule

A conservative computation can now be stated as follows.

1. Declare the experiment, observation resolution, requested functional, and admissible input family.
2. Compute local rank/Lyapunov diagnostics, but do not treat them alone as probability certificates.
3. Propagate several admissible input laws and measure their diameter in more than one task-relevant metric.
4. Sweep observation scale. A conclusion that appears only at one binning remains conditional on that binning.
5. If the diameter falls below resolution after a stable frontier, use Monte Carlo for the resulting forced target and report its numerical validity diagnostics.
6. If the diameter persists, report the input declaration and the range over declarations.
7. At walls, replace a single ellipsoid or interval by the full preimage correspondence.

10 Conclusion

The comparison resolves the apparent competition. Monte Carlo is the correct numerical engine once the target distribution or quotient functional is justified. Uncertainty geometry contributes the missing structural layer. It distinguishes sampling convergence from declaration independence, identifies forecast frontiers, tests robustness across scale, and preserves fibres and branches that a scalar probability report can erase. In the registered regular post-mixing regime, both methods match. Outside that regime, the extra geometric structure is measurable and operational rather than rhetorical.

A Reproducibility

Run:

```
python rg_true_chaos_battery.py --seed 20261010 \
  --outdir rg_true_chaos_output
```

The run writes five PNG figures and `rg_true_chaos_results.json`. The implementation uses NumPy, SciPy, and Matplotlib. Randomness is generated by NumPy’s declared seed. The complete script accompanies this paper.

References

- [1] N. Metropolis and S. Ulam, “The Monte Carlo Method,” *Journal of the American Statistical Association*, 1949.
- [2] E. N. Lorenz, “Deterministic Nonperiodic Flow,” *Journal of the Atmospheric Sciences*, 1963.
- [3] A. Lasota and M. C. Mackey, *Chaos, Fractals, and Noise*, Springer, 1994.
- [4] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, Wiley, 2006.

- [5] C. Villani, *Optimal Transport: Old and New*, Springer, 2009.
- [6] A. W. van der Vaart, *Asymptotic Statistics*, Cambridge University Press, 1998.